

Lab 3: Architecture Selection & Justification

Scenario: Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice – Two Reasons

1. It provides clear separation of concerns between the touch-screen presentation, order/business logic, data access, payment processing, and receipt handling.
2. The kiosk is a focused, self-contained system with defined hardware integrations, so a layered design is simpler to develop, test, understand, and maintain than a microservices architecture.

Security Advantage

Payment processing is isolated in the Payment Service and communicates with the external credit-card gateway through a secure connection. Sensitive payment data can be kept out of the UI and order-management logic, while menu-data access can be restricted to the Menu & Pricing component.

Performance Benefit

Separating menu/pricing access allows frequently used menu data to be cached and reduces repeated database access. Dedicated payment and receipt components also help prevent external operations from blocking the main ordering workflow.

Components & Interfaces

Five required components are shown: Kiosk UI, Order Manager, Payment Service, Menu & Pricing, and Receipt Service. The diagram also shows the Menu Database, Credit Card Gateway, and Receipt Printer. Interfaces include Order Service, Payment Processing, Menu Lookup, Receipt Data, Database Query, Printer Interface, and Secure Payment API.